

Zining Zhu

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🌐 <https://ecailab.org>

🌐 zining-zhu

Employment

2024 - present **Assistant Professor**, Stevens Institute of Technology, Hoboken, New Jersey, US

Employment (Internship)

June 2022 - Sep 2022 **Applied Scientist Intern**, Amazon Search, Palo Alto, California
Advisor: Haoming Jiang, Applied Research Scientist.

Sep 2017 - Aug 2018 **Software Engineering Intern**, Winterlight Labs, Toronto, Ontario, Canada.
Advisor: Jekaterina Novikova, Director of Machine Learning.

June 2017 - Aug 2017 **Software Engineering Intern**, TripAdvisor Inc., Needham, Massachusetts.
Advisor: Raksik Kim, Software Engineer.

May 2016 - Aug 2016 **Research Assistant**, Dynamic Systems Lab, Toronto, Ontario.
Advisor: Angela Schoellig, Assistant Professor.

Education

2019 - 2024 **PhD, University of Toronto** Computer Science.
Advisor: Frank Rudzicz, Associate Professor at University of Toronto. Now Professor at Dalhousie University.
Thesis: *Methods and Applications for Probing Deep Neural Networks*.

2014 - 2019 **BASc., University of Toronto** Engineering Science, Robotics Option.

Grants & Awards

Grants

- *Meta-Capability-Aware Planning for Efficient Tool-Using Agents* NVIDIA Academic Grant Program, 1 DGX Spark ($\approx$ \$4,500), 2026.
- NJFast Plug and Play AI Challenge startup award, \$100,000. 2026.
- *NAIRR: Generating Feasible Follow-up Research Questions*, National Artificial Intelligence Research Resource Pilot Program, National, $\approx$ \$15,000 (in forms of credits and node hours), 2025
- *NAIRR: Accelerating Neural Network Circuits with Software-Hardware Co-Design*, National Artificial Intelligence Research Resource Pilot Program, National, $\approx$ \$15,000 (in forms of computation credits and node hours), 2025
- *Cloud Computing Program Grant*, Google, Institutional, \$5,000, 2025
- *Toward Safer Mixture-of-Expert Models*, Stevens Institute for Artificial Intelligence, Institutional, \$5,000, 2025
- *An LLM-enhanced Framework for Targeted Practices*, Teaching and Learning Center at Stevens, Institutional, \$5,000, 2025
- *Vector Institute PhD Research Grant*, Institutional, \$6,000 each, 2020-2024

Awards

- *Scholera: An AI-Enhanced Learning Platform*, AI in Teaching and Learning Grants Honorable Mention. 2026
- Outstanding paper award, NAACL. 2025
- Top Reviewer Award, NeurIPS. 2023
- Ontario Graduate Scholarship, Provincial, \$15,000. 2022-2023

Publications

Refereed Conference Proceedings

1. Almeida T, **Zhu Z**, and Ning Y. Concept Hierarchies in Large Language Models. *Findings of ACL*. 2026
2. Wang Y, Xu Y, Shen N, Su J, Huang J, and **Zhu Z**. The Illusion of Specialization: Unveiling the Domain-Invariant “Standing Committee” in Mixture-of-Experts Models. *ACL*. 2026
3. Wang H, **Zhu Z**, and Shi HF. Distribution Prompting: Understanding the Expressivity of Language Models Through the Next-Token Distributions They Can Produce. *EMNLP*. 2025
4. Lei Y, Niu J, **Zhu Z**, Chen X, and Penn G. Sheaf Discovery with Joint Computation Graph Pruning and Flexible Granularity. *EMNLP*. 2025
5. Javaji SR, Cao Y, Li H, Yu Y, Muralidhar N, and **Zhu Z**. Can AI Validate Science? Benchmarking LLMs for Scientific Claim $\rightarrow$ Evidence Reasoning. *AACL-IJCNLP*. 2025
6. Javaji SR and **Zhu Z**. What Would You Ask When You First Saw $a^2 + b^2 = c^2$? Evaluating LLM on Curiosity-Driven Questioning. *AACL-IJCNLP*. 2025
7. Li H, Cao Y, Yu Y, Javaji SR, Deng Z, He Y, Jiang Y, **Zhu Z**, Subbalakshmi K, Xiong G, Huang J, Qian L, Peng X, Xie Q, and Suchow JW. INVESTORBENCH: A Benchmark for Financial Decision-Making Tasks with LLM-based Agent. *ACL*. 2025
8. Roewer-Despres F, Feng J, **Zhu Z**, and Rudzicz F. ACCORD: Closing the Commonsense Measurability Gap. *NAACL*. [Outstanding paper award](#). 2025
9. **Zhu Z**, Chen H, Ye X, Lyu VQ, Marasović A, Tan C, and Wiegrefe S. Explanation in the Era of Large Language Models. *NAACL Tutorial Abstracts*. 2024
10. Jingcheng N, Wang A, **Zhu Z**, and Penn G. What does the Knowledge Neuron Thesis Have to do with Knowledge? *ICLR*. [Spotlight \(Top 5%\)](#). 2024
11. Sahak E, **Zhu Z**, and Rudzicz F. A State-Vector Framework for Dataset Effects. *EMNLP*. 2023
12. **Zhu Z**, Shahtalebi S, and Rudzicz F. Predicting fine-tuning performance with probing. *EMNLP*. 2022
13. **Zhu Z**, Wang J, Li B, and Rudzicz F. On the data requirements of probing. *Findings of ACL*. 2022
14. Li B, **Zhu Z**, Thomas G, Rudzicz F, and Xu Y. Neural reality of argument structure constructions. *ACL*. 2022
15. Ramezani A, **Zhu Z**, Rudzicz F, and Xu Y. An unsupervised framework for tracing textual sources of moral change. *Findings of EMNLP*. 2021
16. Li B, **Zhu Z**, Thomas G, Xu Y, and Rudzicz F. How is BERT surprised? Layerwise detection of linguistic anomalies. *ACL-IJCNLP*. 2021
17. **Zhu Z** and Rudzicz F. An information-theoretic view on selecting linguistic probes. *EMNLP*. 2020
18. **Zhu Z**, Novikova J, and Rudzicz F. Detecting cognitive impairments by agreeing on interpretations of linguistic features. *NAACL*. 2019
19. Li Q, Qian J, **Zhu Z**, Bao X, Helwa M, and Schoellig A. Deep neural networks for improved, impromptu trajectory tracking of quadrotors. *ICRA*. 2017

Refereed Workshop Proceedings

20. Desai DA, Huang S, and **Zhu Z**. Distributed Interpretability and Control for Large Language Models. *Annual Meeting of the NAIRR Pilot*. 2026
21. Javaji S, Gauri B, and **Zhu Z**. Another Turn, Better Output? A Turn-Wise Analysis of Iterative LLM Prompting. *NeurIPS MTI-LLM Workshop*. 2025

22. Bhargav S and **Zhu Z**. Feature-Guided SAE Steering for Refusal-Rate Control Using Contrasting Prompts. *NeurIPS Mechanistic Interpretability Workshop*. 2025
23. Li H, Cao Y, Yu Y, Suchow JW, and **Zhu Z**. Truth Neurons. *ACL Knowledgeable Foundational Models Workshop*. 2025
24. Jaipersaud B, **Zhu Z**, Rudzicz F, and Creager E. Show, Don't Tell: Uncovering Implicit Character Portrayal using LLMs. *NeurIPS Workshop on Creativity and GenAI*. 2024
25. Ajwani RD, Javaji SR, Rudzicz F, and **Zhu Z**. LLM-Generated Explanations May be Adversarially Helpful. *NeurIPS Regulatable ML Workshop*. [Featured in Stevens School of Engineering & Science Newsletter](#). 2024
26. Ajwani RD, **Zhu Z**, Rose J, and Rudzicz F. Plug and Play with Prompts: A Prompt Tuning Approach for Controlling Text Generation. *AAAI ReLM*. 2024
27. **Zhu Z**, Jiang H, Yang J, Nag S, Zhang C, Jie H, Gao Y, Rudzicz F, and Yin B. Situated Natural Languages Explanations. *ACL NLRSE*. 2023
28. **Zhu Z**, Shahtalebi S, and Rudzicz F. OOD-Probe: A Neural Explanation of Out-of-Domain Generalizations. *ICML SCIS Workshop*. 2022
29. Shahtalebi S, **Zhu Z**, and Rudzicz F. Out-of-Distribution Failure through the Lens of Labeling Mechanisms. *ICML SCIS Workshop*. 2022
30. **Zhu Z**, Pan C, Abdalla M, and Rudzicz F. Examining the rhetorical capacities of neural language models. *EMNLP BlackboxNLP Workshop*. 2020
31. Hsu YT, **Zhu Z**, Wang CT, Fang SH, Rudzicz F, and Tsao Y. Robustness against the channel effect in pathological voice detection. *NeurIPS ML4H Workshop*. 2018
32. **Zhu Z**, Novikova J, and Rudzicz F. Semi-supervised classification by reaching consensus among modalities. *NeurIPS IRASL Workshop*. 2018

Preprints

33. Mi Z, Tan Q, Yu X, **Zhu Z**, Yuan G, and Huang S. KerZOO: Kernel Function Informed Zeroth-Order Optimization for Accurate and Accelerated LLM Fine-Tuning. 2025
34. Doda S, Javaji SR, and **Zhu Z**. VERBA: Verbalizing Model Differences Using Large Language Models. 2025
35. Vinden N, Saqur R, **Zhu Z**, and Rudzicz F. Contrastive Similarity Learning for Market Forecasting: The ContraSim Framework. 2025
36. Cao Y, Li H, Yu Y, Javaji SR, He Y, Huang J, **Zhu Z**, Xie Q, Liu Xy, Subbalakshmi K, et al. FinAudio: A Benchmark for Audio Large Language Models in Financial Applications. 2025
37. Qiu P, Rudzicz F, and **Zhu Z**. Scenarios and Approaches for Situated Natural Language Explanations. 2024
38. **Zhu Z** and Rudzicz F. Measuring Information in Text Explanations. 2023
39. Huang J, Gao Y, Li Z, Yang J, Song Y, Zhang C, **Zhu Z**, Jiang H, Yin B, and Chang KCC. CCGen: Explainable Complementary Concept Generation in E-Commerce. 2023
40. **Zhu Z**, Balagopalan A, Ghassemi M, and Rudzicz F. Quantifying the Task-Specific Information in Text-Based Classifications. 2021
41. **Zhu Z**, Li B, Xu Y, and Rudzicz F. What do writing features tell us about AI papers? 2021
42. **Zhu Z**, Xu Y, and Rudzicz F. Semantic coordinates analysis reveal language changes in AI research. 2020

43. **Zhu Z**, Novikova J, and Rudzicz F. Deconfounding age effects with fair representation learning when assessing dementia. [Featured in Medical Xpress](#). 2019

Press coverage

- Stevens News: Zining Zhu Helps Developers and Users Navigate ‘Adversarial Helpfulness’ in Artificial Intelligence (December 18, 2024)
- Medical Xpress: A new machine learning model to isolate the effects of age in predicting dementia (July 27, 2018)

Teaching

Instructor

Courses at Stevens Institute of Technology:

- CS 810A Topics in Explainable Natural Language Processing (2025 spring). [New course](#).
- CS 584 Natural Language Processing (2024 fall, 2025 fall, 2026 spring).

Courses at University of Toronto:

- CSC401 / 2511 Natural Language Computing (2023 winter). Co-instructing with En-Shiun Lee and Raeid Saqur
- CSC401 / 2511 Natural Language Computing (2022 winter). Co-instructing with Frank Rudzicz and Raeid Saqur

Teaching Assistant (at University of Toronto)

- CSC108 Introduction to Computer Programming (2023 summer)
- ECE1786 Creative Applications for NLP (course prep TA in 2022 summer and TA in 2022 fall)
- CSC2515 Introduction to Machine Learning (2021 fall)
- CSCC24 Principles of Programming Languages (2021 summer)
- CSC148 Introduction to Computer Science (2021 summer)
- CSC401/2511 Natural Language Computing (2021 winter)
- CSC309 Web Programming (2020 fall)
- CSC401/2511 Natural Language Computing (2020 winter)
- ECE324 Introduction to Machine Intelligence (2019 fall)
- CSC180 Introduction to Computer Programming (2016 fall)

Seminars

- Interpretable NLP seminar at UofT CompLing (2021 winter)
- Introduction to ML seminar at UTADA (2017 fall)

Advising

Current:

- Jikai Long – 2026-present – PhD at Stevens
- Hongbo Liu – 2026-present – PhD at Stevens (co-advised with Steve Yang)
- Ziruo Zhao – 2025-present – PhD at Stevens
- Shashidhar Reddy Javaji – 2024-present – PhD at Stevens
- Preet Jhanglani – 2024-present – PhD at Stevens
- Haohang Li – 2024-present – PhD at Stevens (co-advised with Jordan Suchow).
- Yanyan Xia – 2026 – Master research project. Topic: LM reasoning.
- Tiago Almeida – 2024-present – Undergraduate research project: mechanistic interpretability
- Kevin Doshi – 2025 – Master research project: Scholera: LLM adaptation of practice questions. [Received an NJFast PlugAndPlay AI Challenge Startup Award](#)
- Shijie Zhou – 2026 – Master research project: reinforcement learning
- Colin Brett – 2026 – Undergraduate research project: LM reasoning.
- Jayden Tumbolken – 2026 – Undergraduate research project: Cybersecurity of AI agents.
- Terry Liu – 2025-2026 – Undergraduate research project: Cybersecurity of AI agents.

Alumni:

- Xingyue Qiu – 2025 – Master research project. Topic: Steering vector model steering.
- Manas Satish Singh – 2026 – Master research project. Topic: LM governance.
- Yitao Xu – 2025 – Master research project. Topic: Expert preferences in Mixture-of-expert (MOE)

language models.

- Dev Arpan Desai – 2025 – Master research project (SIAI AIRS program). Topic: Logit lens and steering vectors on multi-GPU LLMs.
- Muhammad Shahzaib Chaudhury – 2025 – Master research project. Topic: Sparse Autoencoders model steering.
- Azizul Haque – 2025 – Master research project
- Molly DiCampli – 2025 Undergraduate research project (Pinnacle Scholar program) Project: Sports performance data analysis.
- Shravan Doda – 2024-present – Master research project. Topic: Natural language explanation of model differences.
- Samaksh Bhargav – 2024 – Edison Academy Magnet School. Topic: Sparse autoencoder (SAE) model steering.
- Rohit Sandadi – 2024 – Dougherty Valley High School. Topic: Natural language explanation of neurons.
- Patrick Wierzbicki – 2024 – Undergraduate research project. Topic: Retrieval language model.
- Paul Gao – 2024 – Undergraduate research project. Topic: Graph-based LLM reasoning.
- Pengshuo Qiu – 2024 – Undergraduate research project: Situated natural language explanations.
- Sean Wang – 2024 – Undergraduate research project. Topic: Analyzing the distribution in next-token prediction.
- Sam Pan – 2024 – Undergraduate research project. Topic: Use steering vector to control model honesty.
- Huaizhi Ge – 2023 – Master research project. Topic: mechanistic interpretability of language models.
- Sanika Mhadgut – 2024 – Master research project. Topic: neural network circuits.
- Jim Yang – 2023 – Summer research project: Natural language explanation
- Jason Zuo – 2023 – Summer research project: Probing and explanation
- Rohan Deepak Ajwani – 2022-2023 – Summer research project & ECE MEng projects. Topic: adversarial helpfulness explanations, controllable generation.
- Philipp Eibl – 2021 – Undergraduate research project: Information estimators
- Esmat Sahak – 2021 – Undergraduate research project: Multitask learning and probing.

Services

Organizing Workshops, Tutorials, etc.

- Explanation in the Era of Large Language Models, tutorial at NAACL 2024
- Machine Learning for Cognitive Mental Health (ML4CMH), workshop at AAAI 2024
- ACL Student Research Seminar at UofT Computational Linguistics, 2023

Reviewing service

- Senior Area Chair: EMNLP (2026)
- Action Editor: ACL Rolling Review (2024-2026)
- Area Chair: ICML (2025-2026), NeurIPS (2024-2026)

Reviewing

- 2026: COLM, ACL Rolling Review
- 2025: ICLR, AAAI, ACL Rolling Review, COLM, LLMSEC@ACL
- 2024: ICLR, ICML, ACL Rolling Review, COLM, AAAI
- 2023: ICLR, ICML, IJCAI, ACL Rolling Review, FAccT, NeurIPS, EMNLP, R2HCAI@AAAI
- 2022: ACL Rolling Review, EMNLP, NeurIPS, RobustSeq@NeurIPS, LT-EDI@ACL, CMCL@ACL
- 2021: ACL, EMNLP, NAACL, AAAI
- 2020: ACL, IEEE Journal of Biomedical and Health Informatics
- 2018: Computer Methods & Programs in Biomedicine

Selected Talks

- *Mechanistic Interpretation and Reasoning AI*, 2nd NJIT Workshop on Multimedia Intelligence (MMI 2026). May 21, 2026.
- *Steering Language Models for Financial Reasoning*, ACL SIG-Fintech x TFAI Webinar. February 1, 2026.
- *LLM, Reasoning, Safety, and Mechanistic Interpretability*, Keynote at Edison Research Symposium. January 17, 2026.

- *Explainable LLM Reasoning for Software*, Keynote at the 1st International Workshop on Explainable Automated Software Engineering. November 20, 2025.
- *Improving LLM Reasoning with Mechanistic Insights*, Keynote at XLLM-Reason-Plan Workshop at COLM. October 10, 2025.
- *Improving LLM Reasoning with Mechanistic Insights*, CUNY Queens, New York. September 15, 2025.
- *Artificial Intelligence for Teaching and Learning*, panelist at Stevens Teaching and Learning Center Symposium. May 15, 2025
- *Mechanistic Interpretability and AI Safety*, USC CSCI699 guest lecture, Online. March 31, 2025.
- *Chatting with GPT about a PDF*, Stevens CS188 guest lecture, March 26, 2025.
- *Mechanistic Interpretability and AI Safety*, Rutgers University New Brunswick, New Jersey. October 15, 2024.
- *A Communication Channel of Explanations*, AI camp event, Toronto. February 7, 2024. Vector Institute NLP Workshop, Toronto. February 16, 2024.
- *Challenge and Opportunities in AI Safety and Alignment*, TGO Virtual Panel Talk. Nov 11, 2023.
- *Towards Interpretable and Controllable AI*, Stevens Institute of Technology (May 18, 2023), UW-Madison iSchool, (February 16, 2023)
- *Towards Interpretable and Controllable AI*, NEC Laboratories Europe, February 2, 2023
- *Interpretable and Controllable Pretrained Language Models*, UT Computational Linguistics talk, Nov 15, 2022
- *Incorporating probing in the development of large language models*, Vector Institute Endless Summer School (ESS) invited talk, March 1, 2022
- *On the data requirements of probing*, Vector Institute Research Symposium, Virtual poster presentation, Feb 22, 2022
- *Predicting fine-tuning performance with probes*, UT Computational Linguistics, virtual presentation, Feb 15, 2022
- *Quantifying the task-specific information in text-based classifications*, UT Language Research Day, Virtual presentation, Nov 12, 2021
- *Probing neural language models*, AISC Recent Trends in NLP discussion, Video talk, Aug 15, 2021
- *Writing can predict AI papers acceptance, but not their impact*, Vector Institute Research Symposium, Virtual poster presentation, Feb 16, 2020
- *Improving the neural NLP model performances with linguistic probes*, Zhi-Yi NLP Open Course, Video talk, Nov 20, 2020
- *An information-theoretic view on selecting linguistic probes*, TsingHua University AI TIME, Video talk, Oct 30, 2020
- *Examining the rhetorical capacities of neural language models*, Vector Institute NLP Symposium spotlight presentation, Video talk, Sep 16, 2020.
- *Speeding up computation with GPU and Google Cloud*, ECE324 tutorial, Toronto, Canada, Oct 31, 2019
- *Efficient pre-training methods for language modeling*, Tencent Jarvis Lab, Shenzhen, China, Aug 5, 2019
- *Automatic assessment of cognitive impairments*, UTMIST tech talk, Toronto, Canada, Nov 20, 2018
- *Probabilistic graphical models*, UTADA tech talk, Toronto, Canada, Oct 21, 2017

Affiliation

Association of Computational Linguistics (ACL), International Electrical and Electronics Engineers (IEEE)